23MID0018

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10. Series Practice – Part 1

a. Create a simple series of Student_Names and Programmes they belong with

In [113…
```
import pandas as pd
data = ["Rajavel","Nideesh","Santhosh"]
index = ["DS","CSE","CSE"]
s = pd.Series(data,index=index)
print(s)
```

```
DS      Rajavel
CSE     Nideesh
CSE    Santhosh
dtype: str
```

b. Create a series from a list, tuple, dictionary and a custom index

In [4]:
```
import pandas as pd
# Series from list
lst = ['a','b','c','d','e']
s_lst = pd.Series(lst,index = [1,2,3,4,5])
print(s_lst)
```

```
1    a
2    b
3    c
4    d
5    e
dtype: str
```

In [115…
```
tup = ('a','b','c','d','e')
s_tup = pd.Series(lst,index = [1,2,3,4,5])
print(s_tup)
```

```
1    a
2    b
3    c
4    d
5    e
dtype: str
```

In [7]:
```
dic = {
    1 :'a',
    2:'b',
    3:'c'
}
s_dic = pd.Series(dic)
print(s_dic)
```

```
1    a
2    b
3    c
dtype: str
```

In [8]:
```
data = ['a','b','c','d']
index =[1,2,3,4]
s_custom = pd.Series(data,index=index)
print(s_custom)
```

```
1    a
2    b
3    c
4    d
dtype: str
```

C. Create two series with CAT I and CAT II marks of students and explore element-wise

operations

In [45]:
```
import pandas as pd
s_cat1 = pd.Series([99,98,76,65],index=["Raj","ram","nideesh","boopeash"])
s_cat2 = pd.Series([100,68,88,98],index=["Raj","ram","nideesh","kishori"])
print(s_cat1 + s_cat2)
print(s_cat1 - s_cat2)
print(s_cat1 * s_cat2)
print(s_cat1 / s_cat2)
```

```
Raj         199.0
boopeash      NaN
kishori       NaN
nideesh     164.0
ram         166.0
dtype: float64
Raj         -1.0
boopeash     NaN
kishori      NaN
nideesh    -12.0
ram         30.0
dtype: float64
Raj         9900.0
boopeash       NaN
kishori        NaN
nideesh     6688.0
ram         6664.0
dtype: float64
Raj         0.990000
boopeash         NaN
kishori          NaN
nideesh     0.863636
ram         1.441176
dtype: float64
```

d.Write conditions to check if data in a series are integers/chars/Not a Number

In [122…

```
import pandas as pd
data = pd.Series([99,98,"null",65],index=["Raj","ram","kishori","boopeash"])
print(data.isna())
print(data.dropna())
print(data.fillna(0))
```

```
Raj         False
ram         False
kishori     False
boopeash    False
dtype: bool
Raj           99
ram           98
kishori     null
boopeash      65
dtype: object
Raj           99
ram           98
kishori     null
boopeash      65
dtype: object
```

e.Create a series for Product Names, Prices, Quantities in a retail store

In [86]:
```
import pandas as pd
data = ["TV","LG",100000,5]
s = pd.Series(data,index=["Product_name","Brand","price","quantity"])
print(s)
```

```
Product_name        TV
Brand               LG
price           100000
quantity             5
dtype: object
```

In [68]:
```
data = [["TV","Laptop","Smart_watch"],[100000,70000,20000]]
index = pd.MultiIndex.from_arrays(data,names = ["Product Names ","Prices"])
s = pd.Series([10,15,7],index=index )
print(s)
```

```
Product Names   Prices
TV              100000    10
Laptop          70000     15
Smart_watch     20000      7
dtype: int64
```

f. Create a Multilevel series with arrays, tuple, product and dataframe

In [78]:
```
import pandas as pd
array = [[1,2,3],["raj","nideesh","kishori"]]
data = [100,88,99]
index  = pd.MultiIndex.from_arrays(array)
s_arr = pd.Series(data,index= index)
print(s_arr)
```

```
1  raj        100
2  nideesh     88
3  kishori     99
dtype: int64
```

In [100…
```
import pandas as pd
array = [(1,"raj"),(2,"nideesh"),(3,"kishori")]
data = [100,88,99]
index  = pd.MultiIndex.from_tuples(array)
s_tup = pd.Series(data,index= index)
print(s_tup)
```

```
1  raj        100
2  nideesh     88
3  kishori     99
dtype: int64
```

In [114…
```
import pandas as pd
pro = [[1,2,3],["raj","nideesh","kishori"]]
data = [100,88,99,78,35,66,67,48,90]
index  = pd.MultiIndex.from_product(pro)
```

```
s_pr = pd.Series(data,index= index)
print(s_pr)
```

```
1  raj        100
   nideesh     88
   kishori     99
2  raj         78
   nideesh     35
   kishori     66
3  raj         67
   nideesh     48
   kishori     90
dtype: int64
```

In [102…

```
values = {
    "Name" : ["Rajavel","Nideesh","Kishori"],
    "Dept":["DS","CSE","CSE"]
}
df = pd.DataFrame(values)
data = [100,88,99]
index  = pd.MultiIndex.from_frame(df)
s_df = pd.Series(data,index= index)
print(s_df)
```

```
Name     Dept
Rajavel  DS      100
Nideesh  CSE      88
Kishori  CSE      99
dtype: int64
```

g.Access the data elements using First Level Index and Second Level Index.

In [103…

```
import pandas as pd
array = [[1,2,3],["raj","nideesh","kishori"]]
data = [100,88,99]
index  = pd.MultiIndex.from_arrays(array)
```

```
s_arr = pd.Series(data,index= index)
print(s_arr[1])
```

```
raj    100
dtype: int64
```

In [106…
```
print(s_arr[2,"nideesh"])
```

```
88
```

h. Access a specific data item using .loc

In [128…
```
import pandas as pd
array = [[1,2,3],["raj","nideesh","kishori"]]
data = [100,88,99]
index  = pd.MultiIndex.from_arrays(array)
s_arr = pd.Series(data,index= index)
s = s_arr.loc[1]
print(s)
```

```
raj    100
dtype: int64
```

In [129…
```
print(s_arr.loc[1:2])
```

```
1  raj        100
2  nideesh     88
dtype: int64
```

d. Write conditions to check if data in a series are integers/chars/Not a Number (continue)

In [126…
```
import pandas as pd
s_cat1 = pd.Series([99,98,76,65],index=["Raj","ram","nideesh","boopeash"])
s_cat2 = pd.Series([100,68,88,98],index=["Raj","ram","nideesh","kishori"])
s = s_cat1 - s_cat2
print(s)
```

```
print("after dropping")
print(s.dropna())
```

```
Raj         -1.0
boopeash     NaN
kishori      NaN
nideesh    -12.0
ram         30.0
dtype: float64
after dropping
Raj        -1.0
nideesh   -12.0
ram        30.0
dtype: float64
```

In [ ]: